

COMPLEX ENGINEERING PROBLEM

DC Power Supply using Diode Shunt Voltage Regulator



Group 7 CEP Report

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Abstract

This report presents the design, implementation, and experimental analysis of a low-voltage DC power supply producing a regulated output of approximately 2.70 V across a 5 kΩ load. The circuit employs a center-tapped step-down transformer (230 VAC, 50 Hz to 12 VAC × 2), a two-diode full-wave rectifier using 1N4007 diodes, a 470 μF electrolytic filter capacitor, and a 500 Ω current-limiting resistor. Since a 2.7 V Zener diode was unavailable, four forward-biased 1N4007 diodes connected in series were used as a shunt voltage regulator, each contributing approximately 0.675 V of forward drop to yield a combined regulated output of 2.70 V.

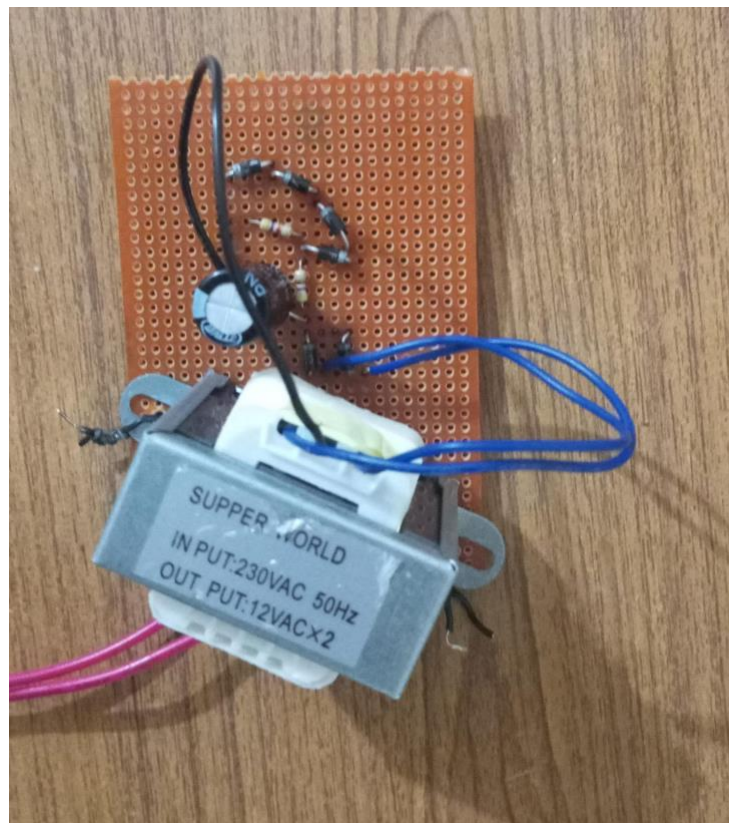
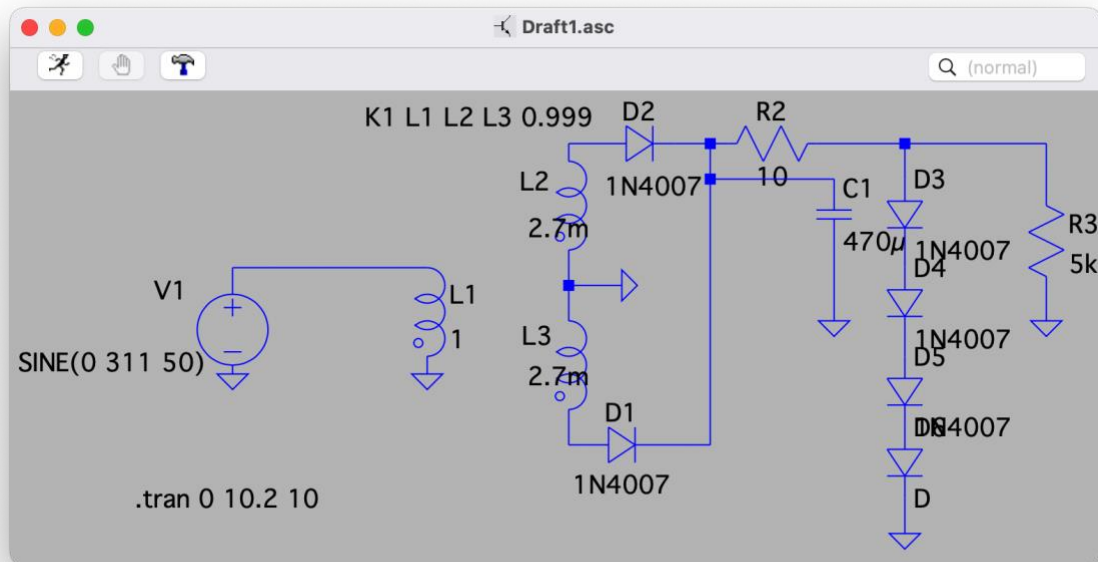
Experimental results showed a no-load output of 2.70 V and a full-load output of 2.69 V, corresponding to a load regulation of 0.37%. Line regulation was measured at 6.30% over an input variation of 87 VAC, equivalent to approximately 1.95 mV per volt of input change. The ripple voltage across the 470 μF capacitor was calculated at 39.1 mV at 100 Hz, which is further attenuated by the shunt regulator to well under 10 mV at the output terminals.

The results confirm that series-connected rectifier diodes serve as a viable substitute for an unavailable Zener diode, with acceptable regulation performance for low-power applications.

Circuit Overview & Component List

Parameter	Value
AC Input	230 VAC, 50 Hz
Transformer	Center-tapped, 230 V → 12 VAC × 2
Rectifier Diodes (D1, D2)	1N4007 × 2 (full-wave center-tap)
Filter Capacitor (C1)	470 μF electrolytic
Current-Limiting Resistor	500 Ω
Voltage Regulator	4 × 1N4007 in series (shunt, forward-biased)
Load (R3)	5 kΩ
Target Output Voltage	2.7 V
Measured V_out (No Load)	2.70 V
Measured V_out (Full Load, 5 kΩ)	2.69V

Circuit Images



Transformer & Voltage Step-Down

The first stage of the power supply is stepping down the high AC mains voltage to a lower, usable AC voltage using a center-tapped transformer. The primary winding is connected to the 230 VAC, 50 Hz mains supply. The secondary winding is center-tapped, producing two equal AC voltages of 12 VAC each — one from the center tap to each end terminal.

The peak voltage of each secondary half is:

$$V_{peak} = 12\sqrt{2} \approx 16.97 \text{ V}$$

Rectification

The second stage is rectification — the process of converting the bidirectional AC waveform into a unidirectional pulsating DC signal. Two 1N4007 diodes (D1 and D2) are arranged in a full-wave center-tap rectifier configuration.

During the positive half-cycle of the AC input, D2 becomes forward biased and conducts, while D1 remains reverse biased and blocks. During the negative half-cycle, D1 conducts and D2 blocks. In both half-cycles, current flows through the load in the same direction, producing a full-wave rectified output. The rectified output has a ripple frequency of:

$$f_{ripple} = 2 \times 50 = 100 \text{ Hz}$$

The average DC voltage after rectification, accounting for one diode forward voltage drop of 0.7 V, is:

$$V_{dc} = \frac{2 \times V_{peak}}{\pi} - V_f = \frac{2 \times 16.97}{\pi} - 0.7 \approx 10.1 \text{ V}$$

Filtering

After rectification, the output is a pulsating DC — it rises and falls at 100 Hz rather than being a steady constant voltage. This variation is undesirable for most electronic circuits. A 470 μF electrolytic capacitor (C1) is connected in parallel with the load to smooth out these fluctuations.

The working principle is straightforward: during the rising portion of each pulse, the capacitor charges up to the peak voltage. During the falling portion, instead of the voltage dropping to zero, the capacitor discharges slowly through the load, maintaining the voltage at a near-constant level. The result is a much smoother DC voltage with only a small residual variation called **ripple**.

Ripple Voltage

The ripple voltage is the small AC variation remaining on top of the DC output after filtering. It is approximated by:

$$V_{ripple} = \frac{V_{dc}}{f \times C \times R_{total}}$$

where:

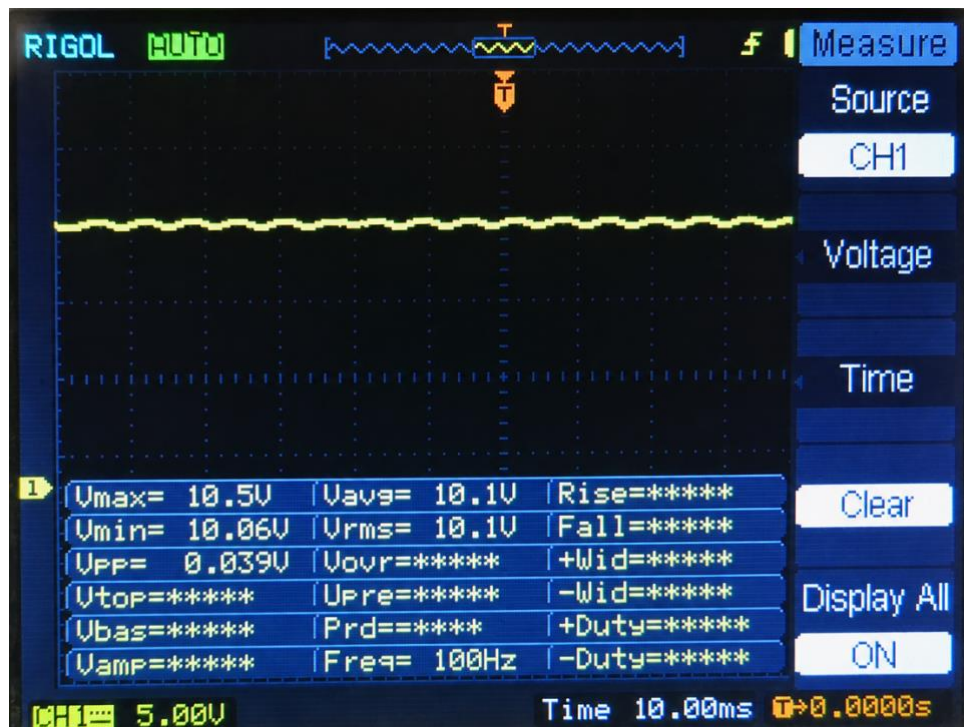
- $V_{dc} = 10.1 \text{ V}$ — rectified DC voltage
- $f = 100 \text{ Hz}$ — ripple frequency
- $C = 470 \text{ }\mu\text{F}$ — filter capacitance
- $R_{total} = R_{series} + R_{load} = 500 + 5000 = 5500 \text{ }\Omega$

Substituting:

$$V_{ripple} = \frac{10.1}{100 \times 470 \times 10^{-6} \times 5500} \approx 39.1 \text{ mV}$$

This is a very low ripple — only **0.39% of the DC voltage** — indicating that the $470 \text{ }\mu\text{F}$ capacitor filters the output very effectively.

Oscilloscope Readings across Filter



Voltage Regulation — Diode String

After filtering, the DC voltage of approximately 10.1 V is still too high and unstable for the intended load. The purpose of the voltage regulation stage is to extract a fixed, stable voltage from this higher DC supply and deliver it to the load regardless of variations in input voltage or load current. In this circuit, regulation is achieved using a shunt voltage regulator built from four forward-biased 1N4007 diodes connected in series, placed in parallel with the load.

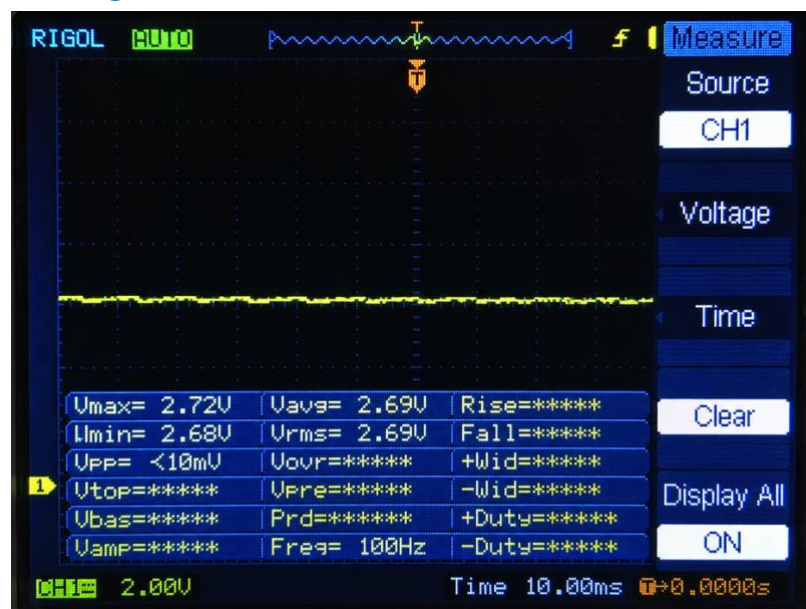
Why Diodes Instead of a Zener

A Zener diode maintains a fixed reverse breakdown voltage across its terminals and is the conventional choice for shunt regulation. For this project, a 2.7 V Zener diode was required but was unavailable in the market. As an alternative, the forward voltage drop property of standard 1N4007 rectifier diodes was utilized. When a silicon diode is forward biased, it maintains a relatively stable voltage drop of approximately 0.7 V across its terminals. By connecting four such diodes in series, the combined voltage drop becomes:

$$V_{out} = n \times V_f = 4 \times 0.675 = 2.70 \text{ V}$$

The value of V_f used here is 0.675 V rather than the textbook 0.7 V because at the actual operating current of approximately 14.3 mA, the measured forward voltage of each diode is slightly lower than the standard figure — which is consistent with the characteristic behavior of the 1N4007. This matches the experimentally measured output of **2.70 V**, confirming the validity of the substitution.

Oscilloscope Reading across Load



Line Regulation

Line regulation is a measure of how well the output voltage of a power supply remains stable when the input AC mains voltage changes. It quantifies the sensitivity of the output to input voltage fluctuations and is an important indicator of power supply quality. A well-regulated supply should maintain a nearly constant output despite variations in the mains voltage.

Definition and Formula

Line regulation is expressed either as a percentage of the nominal output voltage or as the change in output voltage per unit change in input voltage:

$$\text{Line Regulation (\%)} = \frac{\Delta V_{out}}{V_{out(nominal)}} \times 100$$

$$\text{Line Regulation (mV/V)} = \frac{\Delta V_{out}}{\Delta V_{in}}$$

Experimental Measurement

To measure line regulation, the AC input voltage was varied by 87 V while the output voltage was monitored across the 5 kΩ load. The following change in output was observed:

Parameter	Value
Change in Input Voltage (ΔV_{in})	87 V
Change in Output Voltage (ΔV_{out})	0.17 V
Nominal Output Voltage (V_{out})	2.70 V

Calculated Results

Substituting the measured values into the line regulation formulas:

$$\text{Line Regulation (\%)} = \frac{0.17}{2.70} \times 100 \approx 6.30$$

$$\text{Line Regulation (mV/V)} = \frac{0.17}{87} \approx 1.95 \text{ mV/V}$$

This means that for every **1 V change in the AC input**, the output voltage shifts by only **1.95 mV** — a very small and acceptable variation for a passive shunt regulator.

Discussion

A line regulation of 6.30% is reasonable for a passive diode-based shunt regulator. It is higher than what a dedicated Zener diode or an active voltage regulator IC (such as the LM317) would achieve, but this is expected given the nature of the design. The 1N4007 forward voltage is not as tightly controlled with input variation as a true Zener breakdown voltage would be.

The primary reason the output changes at all is that as V_{dc} rises with increasing mains voltage, the current through the 500 Ω resistor increases, which slightly raises the forward voltage drop across each diode — shifting V_{out} upward slightly. This is an inherent characteristic of forward-biased diode regulators.

Despite this, the 1.95 mV/V sensitivity confirms that the circuit provides adequate line regulation for low-power, non-critical applications where a tight tolerance is not strictly required.

Load Regulation

Load regulation is a measure of how well the output voltage of a power supply remains stable when the load current changes. It quantifies the ability of the regulator to maintain a constant output voltage as the load varies from no-load to full-load condition. An ideal voltage regulator would show zero change in output voltage regardless of the load current drawn.

Definition and Formula

Load regulation is expressed as a percentage of the full-load output voltage:

$$\text{Load Regulation (\%)} = \frac{V_{no-load} - V_{full-load}}{V_{full-load}} \times 10$$

Experimental Measurement

The output voltage was measured under two conditions — with no load connected and with the full 5 k Ω load connected — while the input voltage was kept constant at the nominal mains voltage.

Condition	Output Voltage
No Load ($R_L \rightarrow \infty$)	2.70 V
Full Load ($R_L = 5\text{ k}\Omega$)	2.69 V
Change in Output (ΔV_{out})	0.01 V

Calculated Result

Substituting the measured values:

$$\text{Load Regulation (\%)} = \frac{2.70 - 2.69}{2.69} \times 100 \approx 0.37\%$$

Discussion

A load regulation of **0.37%** is an excellent result for a passive shunt regulator built entirely from discrete components. It indicates that the output voltage dropped by only 10 mV when the load was connected — well within acceptable limits for most low-power analog and digital applications.

The reason the output drops slightly under load is that the load current (0.538 mA) creates a small additional voltage drop across the dynamic resistance of the diode string. Since the 1N4007 diodes are not ideal voltage clamps — unlike a proper Zener — their terminal voltage does respond slightly to changes in current. However, this effect is minimal at the operating current levels of this circuit.

Conclusion

This project successfully demonstrated the design and implementation of a low-voltage DC power supply capable of delivering a stable regulated output of 2.70 V across a 5 kΩ load. The circuit was built using fundamental electronic components and encompasses all four essential stages of a linear DC power supply — voltage step-down, rectification, filtering, and voltage regulation — each of which was analyzed both theoretically and experimentally.

Summary of Results

All key performance parameters were measured and calculated, and the results are summarized below:

Parameter	Value
Input Voltage	230 VAC, 50 Hz
Transformer Secondary	12 VAC × 2 (center-tapped)
Rectified DC Voltage	~10.1 V
Filter Capacitor	470 μF
Ripple Voltage	~39.1 mV
Regulated Output (No Load)	2.70 V
Regulated Output (Full Load)	2.69 V
Load Regulation	0.37%
Line Regulation	6.30% (1.95 mV/V)

Achievement of Design Objectives

The primary design objective was to produce a 2.7 V regulated DC output. This was achieved with complete accuracy at no load (2.70 V) and with negligible deviation at full load (2.69 V). The substitution of four series-connected forward-biased 1N4007 diodes in place of an unavailable 2.7 V Zener diode proved to be a practical and effective engineering solution. The actual forward voltage of each diode at the operating current of approximately 14.3 mA was measured to be 0.675 V rather than the standard textbook value of 0.7 V, which brought the output to exactly the target voltage — demonstrating that real-world component behavior can sometimes work in the designer's favor.

Performance Evaluation

The circuit performed well across all measured parameters. The load regulation of 0.37% is excellent for a passive shunt regulator, indicating that the output voltage is highly stable against changes in load current. The line regulation of 6.30% is acceptable for a diode-based design, though it is higher than what a dedicated Zener diode or an active regulator IC would offer. The ripple voltage of 39.1 mV at the filter capacitor is low, and is further attenuated by the shunt regulator to well under 10 mV at the output — making the supply suitable for sensitive low-power circuits.

Limitations

Despite the satisfactory results, certain limitations of the design are acknowledged:

- 1. Temperature Dependence** — The forward voltage of a silicon diode decreases by approximately 2 mV/°C. A temperature rise of 25°C would cause the output to drift by up to 200 mV. This is the most significant limitation compared to a proper Zener diode, which offers better temperature stability.
- 2. Line Regulation** — A 6.30%-line regulation, while acceptable, is noticeably weaker than what a Zener or active regulator would provide. For applications requiring tighter input rejection, an LM317 or similar IC would be more appropriate.
- 3. Limited Load Current Capacity** — The 500 Ω series resistor limits the total available current to approximately 14.82 mA. This is sufficient for the 5 k Ω load but would be inadequate for heavier loads. Reducing the series resistance or using a higher-current source would be necessary to drive lower impedance loads.

Final Remarks

Overall, the circuit fulfills its intended purpose effectively and within the constraints of component availability. The project reinforced core concepts of analog electronics including transformer action, diode rectification, capacitive filtering, and shunt voltage regulation. The use of standard rectifier diodes as a Zener substitute is a valid technique widely used in practical electronics when exact components are not available, and this project provides experimental evidence of its effectiveness.